

Transforming Surgical Care with Extended Reality (XR)

DSO 531 | Vivienne Gunadhi | March 10th, 2026

SECTION 1: TECHNOLOGY AND MARKET OPPORTUNITY

XR Application in Surgical Care

Extended Reality (XR), an umbrella term for Augmented Reality (AR) and Virtual Reality (VR), can overlay digital imagery onto the physical world or immerse users in fully simulated environments. In surgical care, XR devices enable precise, precise, and efficient surgery.

From the early 1990s to the late 2000s, XR devices were mainly used for training and operation planning. XR devices were further integrated into the operation with the release of AR smart glasses in the 2010s. In 2020, neurosurgeons at John Hopkins executed the first AR-guided on surgeries on living patients. The FDA has now recognized more than 100 XR medical devices.

Market Size and Growth

The global surgical care market was valued at **\$743 billion in 2024** and is forecast to reach **\$1.55 trillion by 2035**, driven by aging populations, precise care, and technological advancements. Within this, the XR surgical technology segment is growing with growing clinical evidence.

Key Stakeholders

Stakeholder	Role	XR Relevance
Hospitals / Ambulatory Surgery Centers (ASCs)	Operate ORs; employ surgeons	Primary buyers of XR hardware & software
Medical Device Manufacturers (Medtronic, Stryker, Augmedics, J&J)	Supply surgical tools & platforms	Develop/acquire XR-integrated product lines
Payors / Insurance	Finance & reimburse procedures	Benefit from improved operation outcomes
CMS / FDA	Regulate devices & payment codes	Clearance pace and Current Procedural Terminology (CPT) codes shape adoption speed
Patients	Receive surgical care	Benefit from shorter ops, fewer complications, better rehab

Table 1. Key stakeholders in the surgical XR ecosystem.

SECTION 2: COST STRUCTURE AND COMPETITIVE POSITIONING

Traditional Surgery Cost Drivers

Surgical costs are dominated by time: OR time is billed per minute, meaning a one-hour procedure could incur a \$1,320 to \$7,980 facility cost before supplies, staff, or anesthesia. Non-billable supplies – gloves, packs, sutures, and consumables - add an estimated \$150 to \$210 per hour, which does not include specialty equipment. Basic staff cost will incur at least \$73 per hour in base wages alone. Anesthesia with staffing and supply costs may average \$8.62 per minute, or approximately \$517 per hour of case time.

XR vs. Robotic Systems - Cost and Performance Comparison

XR devices and robotic surgical systems both reduce complications and OR time, but can differ sharply regarding other costs:

Dimension	XR Devices	Robotic Systems (e.g., da Vinci)
Upfront Hardware Cost	\$200 – \$200,000	\$1 – \$2.5 million
OR Time Reduction	10 – 15%	10 – 20%
Post-Op Complication Reduction	10 – 20%	Up to 30%
Training Cost Reduction	30 – 40%	Marginal / increases
Cadaver Replacement	Yes (\$3k–5k/cadaver eliminated)	No
Best Fit	Cost-constrained hospitals and ASCs, training programs	Hospitals and ASCs, academic programs

Table 2. XR vs. robotic surgical systems: cost and performance comparison.

SECTION 3: FUTURE SCENARIOS

Three distinct futures are plausible based on different key drivers. Different market segments may follow these paths simultaneously”

	Scenario 1: Efficiency-Driven (Mid Integration)	Scenario 2: Patient-Centered (High Integration)	Scenario 3: Training-First (Limited Integration)
Key Driver	Hospitals prioritize OR throughput; time savings without more staff	Payer reimbursement expands to cover XR-guided procedures; outcomes data matures; improvement of operation efficiency	High capital cost blocks OR adoption; hospitals invest only in training use cases
Revenue Assumption	Savings from reduced OR time offset device cost in 2–3 years; device pricing decreases as competitors enter	New CPT codes; Scenario 1 assumptions	SaaS-based training licenses (\$5k–\$30k/yr per institution) replace one-time hardware sales as primary model
Who Wins	Medical device manufacturers	SaaS platforms (training + rehab apps); new entrants offering outcomes-as-a-service	Dedicated XR training companies; telemedicine/SaaS platforms; remote proctoring services
Who Is Disrupted	Robotic system vendors lose share in cost-sensitive segments	N/A	High-cost device makers with no software pivot; traditional medical education companies (cadaver, mannequin suppliers)
Critical Risk	Slow hospital IT integration; surgeon resistance to learning curve	Reimbursement timeline uncertain; FDA pathway for patient-facing XR remains complex	Hospitals delay even training investment; further approval by medical board

Table 3. Scenario matrix — key drivers, revenue assumptions, winners and risks.

SECTION 4: STRATEGIC AND BUSINESS IMPLICATIONS

Stakeholder	Strategic Opportunity	Recommended Action
Device Manufacturers	XR expands addressable market beyond robotic systems	Further develop XR software, or acquire or partner with XR software platforms before SaaS players build clinical data moats
Hospitals & ASCs	Reduce OR time, complication rates, and training costs with lower upfront investment than robotics	Integrate XR training and XR into the OR based on surgery volume and related clinical data. Look into establishment of XR usage for patients.
SaaS & Digital Health	New avenues for license revenue potential in training and rehab	Partner with device manufacturers and academic medical centers early to build clinical data. Target training platforms and patient care.

Table 4. Strategic implications by stakeholder group.

Final Recommendations: XR is deployable today with a quantifiable ROI case and a clear regulatory pathway. Companies that establish clinical evidence and EHR integrations can greatly strengthen their position in this competition.

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